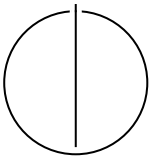
SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Techniques for Using Neural Architecture Search in Federated Learning

Max Coetzee



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Techniques for Using Neural Architecture Search in Federated Learning

Techniken um Neurale Architektursuche für Föderatives Lernen zu nutzen

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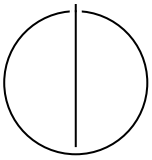

I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 30.07.2026 Max Coetzee

Acknowledgments

Abstract

Both *Neural Architecture Search* (NAS) and *Federated Learning* (FL) have made significant progress independently in the past decade. To benefit from the advantages of NAS methods in FL, researchers have started combining them by using NAS in FL [16] [9] [29].

[TODO: overhaul]

Applying techniques from Neural Architecture Search (NAS) to Federated Learning (FL) has been fruitful (remove) in recent years. The combination was identified as a promising research (remove) direction by [22]. It has yielded methods for finding architectures that deal with the challenges imposed by the FL setting.

Research into NAS has grown rapidly [55] since it was popularized by [79]; consequently, literature on its application to FL has grown. The last survey on NAS applied to FL compared approaches of four papers [78]. Since then, we have identified approximately 50 new papers. This motivates a new systematic survey of the landscape to identify progress and gaps in the literature.

In this thesis, we propose a map of the literature landscape based on the FL challenges they address. We achieve this by systematically evaluating the literature and identifying which challenge it solves.

We refer to the FL challenges described in [79], i.e., non-IID data, limited communication, client heterogeneity, privacy of client data, and break them down into smaller subchallenges — each subchallenge being associated with a pattern in the literature. We include personalized FL [48] as an additional subchallenge that was not originally posited, but has since drawn the community's attention.

We then analyze how the subchallenges are addressed and focus on the contribution of the used NAS method towards overcoming the subchallenge. For each subchallenge, we keep track of the NAS types used (following [55], [2]) and assess whether the underexplored methods are candidates for future research.

Neural Architecture Search

Contents

1 Introduction

Deep learning has seen widespread adoption over the last decade and is being applied to an increasingly diverse set of tasks. Applying deep learning traditionally involves a team of deep learning and domain experts who tailor a neural network architecture to a specific task through a lengthy process of trial and error. To automate this process, researchers invented *Neural Architecture Search* (NAS) [11] methods. NAS methods employ diverse strategies to automatically search for a neural network architecture for a given deep learning task within an architecture search space.

Independently, but in parallel to NAS, researchers developed a machine learning approach called *Federated Learning* (FL) [35], which enables training an ML model on data distributed across a set of clients without sharing their data. The model is distributed from the server to the clients, which train the model on their local data. The clients then send the model weight updates to the server, which aggregates them and redistributes them to the clients. This process is repeated for several rounds. FL enhances the privacy of client data and enables training even when clients' data is too large to transfer effectively to a central training site or cannot be shared due to regulatory or privacy constraints.

There has been significant interest in using NAS in FL over the last couple of years, spawning *NAS-in-FL methods* [17], but using NAS methods in FL is not straightforward. Most NAS methods were developed for a centralised setting and can not be applied directly in FL. For example, in a centralised setting, NAS methods typically run on worker nodes connected via low-latency, high-bandwidth links. However, in FL, a NAS method may run on clients connected via an unreliable, high-latency, low-bandwidth network. This creates a *challenge* when transferring the weights of large candidate models between clients and the server to coordinate an architecture search. Specialized solutions are needed to overcome such challenges for NAS-in-FL methods, such as only transferring encodings of candidate architectures to clients [45].

The NAS-in-FL literature proposes many solutions to address these challenges, but their mechanisms are often obscured because they are bundled as a part of a NAS-in-FL method. Earlier reviews classify NAS-in-FL methods by properties such as when search occurs and which objectives it optimizes [78], summarize selected methods within broader reviews [24], or discuss NAS as part of multi-objective FL [15]. More recent work supports a more fine-grained comparison through a generic NAS-in-FL pipeline, design elements, and method archetypes [17]. These contributions organize whole NAS-in-FL methods and their configurations, but recurring solution mechanism and the challenges they address remain scattered across the literature.

Examining the mechanisms of solutions to challenges individually would make it possible to assess their potential for reuse in new NAS-in-FL methods, but solution mechanisms exist at different levels of abstraction and use different terminology throughout the literature. A synthesis of recurring challenges in the NAS-in-FL literature and their solution mechanisms is needed as well as an understanding of their applicability across NAS-in-FL methods. This leads us to our research question:

Can solution mechanisms identified in the NAS-in-FL literature be re-used?

To answer our research question, we first extract a coherent set of challenges and their related solutions and arrange them around central challenge themes.

We propose grouping challenges and solution around central themes (or should it be around which parts of the NAS-in-FL scenario activates it?) recurring combinations of FL system characteristics and NAS method types which we call NAS-in-FL scenarios. NAS-in-FL scenario is part of my contribution.

To tackle our research question, we conduct an extensive systematic literature review. We collect an

initial set of relevant literature with a Scopus [10] search and extend it with a snowball search. We first use open coding to identify, from the literature, assumption discrepancies and resulting challenges, when using NAS in FL. Next, we extract FL system characteristics and use axial coding to code the influence of characteristics on the relevance of challenges. Then, we extract techniques through open coding and iteratively refine them to synthesise a coherent set of mutually exclusive and collectively exhaustive techniques. Finally, we use axial coding to map techniques onto challenges they overcome.

Our review aims to synthesise techniques for using NAS in FL from the literature to make finding suitable techniques easy in practice. We make the following three contributions. First, by providing a consolidated overview of the challenges of using NAS in FL, practitioners can grasp which issues they can expect to encounter at a glance. Second, by describing how FL system characteristics influence the relevance of challenges, we enable practitioners to focus on the challenges relevant to them. Third, by extracting techniques for using NAS in FL and highlighting the challenges they address, we enable practitioners to compare existing techniques for using NAS in FL for their specific set of challenges.

In Chapter **??**, we cover the background required for this thesis and related work. In Chapter 3, we describe the method with which we perform our literature review and give an overview of the included literature. In Chapter **??**, we highlight the challenges with FedNAS and how they arise from different FL settings. In Chapter **??**, we describe the adaptation techniques we synthesised and how they overcome challenges. In Chapter 7, we conduct a discussion about our work. Chapter 8 contains our conclusion.

2 Background

2.1 Neural Architecture Search

Neural architecture search (NAS) automates the design of neural network architectures. A NAS method consists of a search space, a search strategy and a performance-estimation strategy [11]. The search space determines the potential candidate architectures, and the search strategy decides which candidates to evaluate next based on the performance-estimation strategy.

2.1.1 NAS Method Type

We define a NAS method type as a triple comprising a search-strategy type, a search-space type and a performance-estimation type [11, 55].

We slightly modify the categorisation of and group *search strategies* into black-box optimization, supernet-based methods and other strategies. Black-box optimization includes reinforcement learning, evolutionary and genetic algorithms, Bayesian optimization and Monte Carlo tree search. Supernet-based methods are differentiable when gradients optimize continuous architecture parameters and non-differentiable when discrete subnetworks are selected without such parameters.

Search spaces can be cell-based, hierarchical, chain-structured or other. Cell-based search spaces search reusable computational blocks, hierarchical spaces compose structures across multiple levels and chain-structured spaces vary a sequential series of layers.

Performance estimation is classified as observed validation performance after brief training, surrogate-predicted performance or another estimator. Brief training is the predominant approach in the reviewed NAS-in-FL literature.

2.2 Federated Learning

Federated learning (FL) enables clients to train a model collaboratively while keeping their raw data local. A server distributes a global model to selected clients, which train it locally and return updates for aggregation [35]. Cross-device FL generally involves many resource-constrained and intermittently available devices, while cross-silo FL involves fewer, more reliable organizations. Data may also be partitioned horizontally by samples or vertically by features. These system characteristics shape the resources, coordination and trust available to the federated process [22].

2.3 NAS in FL

NAS in FL performs an architecture search through the federated process without centralizing client data. Candidate evaluation is performed on the client, while candidate generation, evaluation, search-state updates and selection may occur at the server, the clients or both [17]. Depending on the search strategy, clients exchange model weights and search state for separate candidates or a shared supernet. This adds resource and coordination demands to fixed-model FL. The applicability of a challenge or solution therefore depends on both the FL system characteristics and the NAS design.

2.3.1 NAS-in-FL Scenario

We introduce the term *NAS-in-FL scenario* for a tuple comprising a NAS method type and a set of FL system characteristics based on [22, 6]. *Data heterogeneity* distinguishes homogeneous from heterogeneous client distributions, while *data balance* captures whether clients hold balanced or imbalanced quantities of data. *Client computation resources* may have homogeneous or heterogeneous capacities, and the *network environment* may provide homogeneous or heterogeneous communication conditions. The *federation environment* describes whether client availability is stable or dynamic, and the *deployment environment* may impose uniform or heterogeneous requirements. Finally, *trust and privacy requirements* may demand only data locality or additional guarantees for confidentiality and integrity.

3 Method

We conduct a literature review following [52] and analyse the literature on the basis of thematic analysis to extract challenges and solutions for using NAS in FL from the NAS-in-FL literature.

3.1 Literature Selection

We start our literature selection with a search across titles, abstracts and keywords of English-language publications indexed by Scopus using the query `"federated learning" AND "neural architecture search"` [10]. A publication is eligible when it describes a method in which NAS is performed in FL. We excluded publications that address only NAS or FL independently and publications in which centrally completed NAS merely preceeds FL without NAS. Literature reviews are excluded from the analysis and discussed in related work.

For inclusion in our thematic analysis, a publication must provide evidence for at least one challenge or solution. A challenge meets this threshold when the publication connects a condition of FL to a consequence for architecture search. A solution meets the threshold when the described procedure contains a deliberate intervention whose effect on the NAS-in-FL process can be identified. General claims that an approach addresses concerns such as heterogeneity, efficiency or privacy without a suffienctly detailed explanation do not meet this threshold.

We extend this initial set of literature through backward and forward citation searching of the eligible publications. Newly identified publications are screened using the same criteria, and citation searching ends when an iteration adds no additional eligible publications.

3.2 Literature Analysis

We analyzed the literature in two stages. First, we used thematic analysis to develop challenge and solution themes. Second, we assessed the conditions under which each solution mechanism could be reused in another NAS-in-FL scenario.

3.2.1 Extracting Challenges and Solutions

We read every eligible publication for familiarization and coded relevant passages inductively following [3] and used a concept matrix for an initial coding [52]. For each publication, we recorded extracts that code challenges, solutions, the NAS method type, and the targeted FL system characteristics. We recorded the source and page for each extract. A challenge code expressed a condition-consequence relationship specific to performing NAS in FL. A solution code expressed a mechanism for extending the NAS-in-FL method.

We grouped challenge codes when they described the same consequence for architecture search and grouped solution codes when their mechanisms were conceptually similar. Similar terminology alone did not justify grouping. We retained a challenge when it made one coherent analytical claim and a solution when it contained one independently deployable mechanism. General NAS or FL concerns without a distinct consequence for NAS-in-FL were discarded.

We followed the general stages of thematic analysis summarized in Table 3.1 [3]. The process was iterative: while reviewing themes for internal homogeneity and external heterogeneity, we returned to the coded extracts and the corpus as needed. We merged contextual duplicates, split independent claims or mechanisms, and clarified boundaries between related themes. Refinement ended when another review identified no justified structural merger, split, or discard.

Table 3.1: General stages of thematic analysis.

Stage	**Main activity**	**Output**
Familiarization	Read and reread the data and record initial analytical observations.	Familiarity with the corpus and preliminary notes.
Initial coding	Systematically label features of the data that are relevant to the research question.	A set of codes linked to data extracts.
Generating themes	Group related codes and extracts into candidate patterns of shared meaning.	Candidate themes and an initial theme map.
Reviewing themes	Assess the coherence of each theme and the distinctions between themes against the extracts and the complete corpus.	A revised set of themes and theme structure.
Defining and naming themes	Clarify the scope, central idea, and boundaries of each theme.	Concise theme definitions and names.
Reporting	Select supporting extracts and integrate the thematic interpretation into the research account.	A coherent analytical narrative.

Finally, we linked a solution to a challenge when a publication explicitly presented the intervention as a response or when its documented mechanism supplied a traceable effect chain. Co-occurrence within one publication was insufficient. We also recorded applicable conditions, secondary benefits, counterexamples, and trade-offs for the subsequent reuse analysis.

3.3 Determining Solution Mechanism Re-usability

4 Related Work

Existing NAS-in-FL research addresses six broad purposes: coordinating federated search, reducing system costs, differentiating client architectures, protecting the search process, extending the optimization setting, and applies NAS in FL to specific domains.

General search-process research distributes candidate training and evaluation across clients while coordinating the search at the server [14, 41, 31, 16, 64, 77]. These publications establish complete NAS-in-FL methods that often combine several coordination mechanisms.

Resource-oriented research reduces search and deployment costs using methods tailored to computational, communication, and device constraints [67, 68, 49, 53, 75, 9, 65, 29, 37, 72, 66, 25, 4]. Their applicability depends on which cost is constrained.

Personalization research searches architectures at different levels of the federation and coordinates heterogeneous client models [71, 34, 74, 18, 28, 62, 60, 61, 63, 36]. This work separates the goal of differentiated architectures from the problem of coordinating them.

Security research uses privacy-preserving and robust mechanisms to protect search information and models [46, 69, 30, 39, 70, 73]. The available protection mechanisms depend on the representation exchanged during search.

Further work extends NAS-in-FL to broader objectives, model classes, and federated learning settings [45, 38, 43, 19, 54, 50, 51, 76, 20, 12, 33]. These extensions introduce assumptions that restrict transfer to other settings.

Application research evaluates NAS-in-FL across a range of specialized domains [23, 32, 40, 47, 58, 57, 5, 44, 27, 59, 56, 42, 26, 7, 8, 21]. It demonstrates domain breadth, although specialized evaluations provide limited evidence about mechanism reuse elsewhere.

4.1 Prior Syntheses of NAS-in-FL

One early review classifies complete methods by online and offline execution or by single- and multi-objective search [78]. Broader reviews compare NAS with hyperparameter optimization or place architecture search within multi-objective FL [24, 15]. These classifications organize procedures and objectives rather than the relationship between challenge and solution.

The most similar synthesis decomposes NAS-in-FL into design elements, method archetypes, traits, and suitable FL systems [17]. General NAS and FL reviews supply complementary concepts for search design and federated constraints [11, 1]. Their units remain complete methods, NAS components, or FL techniques.

4.2 Position of This Review

This thesis instead synthesizes recurring challenges and solution mechanisms. It separates solution mechanisms that are part of complete NAS-in-FL methods and maps their many-to-many relationships with challenges. This perspective supports assessing reuse in a specified NAS-in-FL scenario without claiming that unevaluated combinations will work.

5 Challenges and Solutions for Using NAS in FL

Our thematic analysis revealed ten key challenges acting as sub-themes of five themes relating to FL system characteristics: *client resource constraints*, *client heterogeneity*, *decentralized search state*, *participation dynamics*, and *security*.

A challenge may have several solutions, and one solution may address several challenges. For clarity, each solution appears under the challenge it primarily addresses. Links to other challenges are identified as cross-cutting. We generally qualify challenges and solutions with the NAS method types of the publications they orignate from.

5.1 Client Resource Constraints

Training many candidate architectures from scratch makes NAS computationally demanding [55]. FL clients often have much less compute and memory available than their centralized counterparts. The challenges below deal with this difficulty.

Challenge 1: Client Capacity Limits the Search Work Assigned Per Round

NAS-in-FL becomes infeasible when an assigned search workload exceeds a client's compute or memory. Hardware differences compound this problem because the same workload may fit one client but not another [9, 68, 14, 61, 4]. Differentiable supernet-based strategies are particularly demanding because clients update several candidate operations within a shared supernet [16, 46, 43, 57]. Other strategies usually assign one candidate at a time, but even that candidate can exceed a low-end client's capacity. This challenge concerns whether clients can execute their assigned task. Challenge 9 concerns how long capable clients take to complete it.

Solution 1a: Use resource-efficient search components. Practitioners can restrict the search space to inexpensive components, such as depthwise-separable convolutions and mobile-oriented CNN blocks [57, 44]. This makes every candidate cheaper to train, but may exclude costly components that improve prediction accuracy. Solution 10a instead tests candidate modules on server data.

Solution 1b: Use capacity-compatible subnets or paths. Rather than make every client train the complete supernet, the search can assign a subnet or active path that fits the client's limits. Its depth, width, channel count or sampling budget can reflect available compute and memory [9, 49, 61, 65, 4, 63]. Single-path local search provides a related way to reduce memory use [14, 53, 27]. As a result, some operations or widths may receive fewer updates, as discussed in Challenge 6.

Solution 1c: Search trainable adaptation modules over a frozen backbone. When a suitable pretrained model is available, clients can keep its backbone fixed and search for a small combination of lightweight adaptation modules. Only these modules and their architecture parameters require local training and exchange, reducing computation and update size [54]. The benefit depends on how well the fixed backbone transfers to the clients' tasks.

Cross-cutting solution 7d. Architecture-agnostic distillation lets small client models teach a server-held supernet without requiring clients to run it [62].

Challenge 2: Architecture Search Adds Computational Cost

Even when clients can handle each round, the complete architecture search can take too long. Black-box strategies may train several candidates in separate federated evaluations [67, 77]. Weight-sharing strategies instead repeatedly update a large supernet or sampled subnets [16, 25]. Split-model execution can also leave clients idle while other segments run [20]. Some strategies discard the search-trained weights and retrain the selected architecture from a new initialization, adding more local epochs and aggregation rounds [59, 73]. This challenge arises from repeated candidate evaluation, idle resources or search work that produces no reusable weights. Challenge 3 addresses communication volume.

Solution 2a: Use multi-fidelity evaluation for candidate architectures. Black-box strategies can first evaluate each candidate with only a few communication rounds. Intermediate validation performance determines which candidates stop early and which receive more rounds. This multi-fidelity approach is used across several black-box search strategies and follows established AutoML methods [13].

For example, [67, 68] report that the best candidate usually becomes distinguishable within the first few rounds. Their procedures drop weaker candidates and increase the budget in later search iterations. Some evolutionary strategies likewise use a small fixed budget or early stopping [76, 51].

Solution 2b: Prune a supernet progressively. Supernet-based strategies can periodically remove weak operations, leaving fewer alternatives to store and train in later rounds [14, 61]. Unlike changing the evaluation budget, pruning reduces the search space itself.

Whether the final model needs separate training depends on how its weights are retained, as addressed by Solution 2e and Solution 2g. Pruning saves more work as search progresses, but early rounds still train the complete supernet and premature pruning can remove useful operations.

Solution 2c: Predict candidate performance. Performance prediction reduces costly configuration evaluations in AutoML [13]. NAS-in-FL strategies can evaluate a small set of architectures and train a predictor to estimate the quality of the rest [62, 47].

Because private, non-IID data can produce different rankings at each client, a separate predictor can be trained for each client [62]. Predictors can also be learned collaboratively without transmitting locally evaluated architectures [31]. This reduces client evaluations, but works only if the evaluated architectures represent the relevant search-space regions.

Solution 2d: Evaluate independent candidates or subnets in parallel. Black-box strategies can train and evaluate different candidates concurrently on separate client groups [67, 77, 47].

Supernet-based strategies can sample several subnets, assign each to a client group and merge their updates into the shared supernet [29]. This can reduce search time when enough clients are available, but different group data can bias candidate comparisons [67]. Challenge 6 addresses this problem.

Solution 2e: Reuse search-trained weights for candidate evaluation and deployment. In weight-sharing strategies, candidates can inherit the operation weights associated with their architecture [50, 72]. This avoids training every candidate from scratch and can initialize the selected subnet before limited fine-tuning [59]. Here, architecture selection remains separate from supernet training. Solution 2g instead produces the architecture and deployable weights together. Shared weights may still rank candidates poorly or provide a weak final initialization.

Solution 2f: Use split-training idle periods for local search. Temporary input and output components can make an assigned model segment executable on its own. During idle periods, a client can then sample and train alternative branches without waiting for the other model segments [20]. The assigned segment can also provide the feature-based teaching signal described in Solution 7d. This converts waiting time into search work.

Solution 2g: Jointly derive an architecture and its deployable weights. One-stage supernet strategies can train model weights while narrowing the architecture choices until the search produces a discrete model with trained weights [14, 19]. Reinforcement-learning strategies over a shared supernet can similarly use later rounds to fine-tune the final model as sampling concentrates on one path [63]. Unlike Solution 2b,

this makes final model production part of search. The selected path may remain undertrained if competing paths received most updates.

Challenge 3: Architecture Search Increases Communication Costs

FL with a predefined architecture exchanges one model's parameters. NAS-in-FL may send larger updates, synchronize more often and exchange many candidate models. A supernet stores weights for alternative operations and can be much larger than the selected architecture. Differentiable strategies may also send architecture parameters throughout search [9, 62, 16, 5].

Solution 3a: Use compact codes to identify known candidates. When clients already hold a compatible master model or trained supernet, the server can send a compact selector or genotype instead of the candidate's parameters. Clients reconstruct the subnet locally and return its evaluation or updated weights [77, 59, 44]. Unlike Solution 7b, this requires a shared search space and common retained weights.

Solution 3b: Synchronize architecture parameters less frequently. Clients can update and upload architecture parameters less often than model weights. One approach transmits them every H_α rounds and stops once the architecture is fixed, while weight training continues [5].

Solution 3c: Aggregate search updates hierarchically. A nearby aggregator can combine updates from several clients before forwarding one result to the remote server. This shifts most aggregation to local links and reduces the search traffic transferred over the wider network [29].

The selected architecture also determines later communication costs. A model chosen only for predictive performance or local computation may have many parameters that FL must transmit in every training round. It can therefore meet compute and memory limits while exceeding the transfer budget [76]. This challenge covers communication during search and during later training of the selected model.

Cross-cutting solution 4b. This solution can include model size or expected transmitted bytes in candidate scores, allowing the search to consider later communication costs before selecting an architecture [76, 5].

5.2 Client Heterogeneity

Clients can differ substantially in compute, network capacity and data distributions. These differences affect which architectures are suitable and how candidate updates should be interpreted.

Challenge 4: No Single Architecture Meets All Deployment Requirements

A NAS-in-FL deployment may include devices with different inference-time compute, memory and latency limits. An architecture sized for the weakest client can underuse stronger hardware, while one sized for the strongest client may not run elsewhere [9]. The search must therefore produce architectures for several targets. Running an independent federated search for each target repeats computation and communication, so costs grow with the number of device groups or deployment budgets [25].

Solution 4a: Train one supernet for different deployment constraints. A weight-sharing supernet can train subnets with different depths, widths or exits in one federated process. Clients with similar limits can receive one architecture per device group [9, 71]. Finer-grained selection can instead choose a subnet for each client's hardware limits and validation data [25, 62, 74].

This solution selects the deployed model, while Solution 1b adjusts the workload a client executes during search. In [25], local predictor can select subnets without further federated training, reducing training cost by a factor of 11 for 20 targets in one evaluation, but training many subnets together can cause interference and leave rarely sampled parameters stale.

Solution 4b: Include resource costs in the architecture search objective. Supernet-based strategies can score predictive performance together with measured deployment costs. Latency or operation-count

limits can tailor the result to each client group's hardware [71, 27, 65, 70]. The score can also penalize model size or expected bytes transmitted during later federated training [76, 5]. Unlike Solution 1a and Solution 10a, this changes how candidates are scored without removing them. The search can therefore balance prediction quality, inference limits and later communication before selecting an architecture.

Challenge 5: Data Heterogeneity Creates Conflicting Architecture Updates

Architecture updates depend on each client's data. Differences in class balance and features can make clients favor different operations or rank the same candidates differently [34, 25]. Clients solving related tasks may also require different outputs [18]. Architecture preferences can also vary over time as client data regimes change [5].

Solution 5a: Cluster clients by learned architecture preference. Clients with similar locally learned architecture parameters can form a group and train one architecture together [40]. Unlike grouping by latency or data balance, this mechanism groups clients by their learned architecture preferences.

Solution 5b: Optimize the worst-performing subnets across client groups. Instead of minimizing only average loss, the search can focus on the subnet with the highest loss in each client group. Sampling the smallest, largest and random subnets can approximate this objective [25]. Giving more attention to poorly performing client-subnet pairs prevents easier pairs from dominating shared-weight training.

Solution 5c: Split the architecture into shared and task-specific components. When clients solve related tasks, the search space can separate a shared feature extractor from task-specific outputs. Clients collaborate on the shared structure while selecting and fine-tuning local components for their task [18].

Cross-cutting solution 6b. Grouping clients to improve data coverage can make each candidate's evaluation better reflect the target population [29].

Challenge 6: Capability-Based Assignment Biases Candidate Evaluation

Assigning only subnets that fit each client's hardware allows more clients to contribute, but also determines which parts of the supernet they train. Operations and widths found only in large subnets may receive updates from few capable clients, while unselected parameters receive none [9, 61, 4]. Treating all parameters as equally trained can then distort the shared search state.

With non-IID data, assignment also ties each model size to the data of clients that can run it. Small candidates may train mainly on low-capability clients and large candidates on well-connected clients [9]. Candidate scores may then reflect the assignment pattern rather than architecture quality, even when every selected client returns an update [62]. Unequal training and biased data coverage are separate effects of the same assignment process.

Solution 6a: Aggregate each parameter only from clients that trained it. The server can aggregate an operation only from clients that updated it and weight their updates by the number of examples processed by that operation [9, 65, 4]. The resulting aggregation weights differ by parameter and reflect how much training each received. This avoids treating missing parameter updates as ordinary client contributions.

Solution 6b: Form groups based on resources and representative data. Client groups can meet a time or bandwidth limit while jointly representing the target class distribution. Methods either rebalance bandwidth-based groups by class distribution or minimize each group's difference from the population under a completion-time limit [37, 29]. Forming a group for each candidate makes comparisons less dependent on which clients can run it. The same approach can keep recurring client availability from biasing round-level participation.

Cross-cutting solution 7d. Architecture-agnostic distillation lets candidates learn from client predictions or features even when those clients cannot run the candidates [62, 34].

5.3 Decentralized Search State

NAS-in-FL distributes architecture parameters and model weights across clients and rounds. The resulting challenges concern whether these updates can be combined meaningfully and whether candidate evaluations remain comparable as the FL setup and architecture change.

Challenge 7: Distributed Search States Cannot Always Be Averaged

Differentiable NAS trains model weights and architecture parameters together in a shared supernet. The architecture parameters control which operations contribute, and their updates depend on the current operation weights. Aggregating only model weights loses local architecture decisions. Combining architecture parameters with unrelated weights separates those decisions from the state that produced them [16, 14, 43]. Direct aggregation therefore assumes that every client uses the same supernet and that its variables have the same meaning.

Independent local searches can return graphs with different connections, operations or blocks. Because their variables refer to different structures, ordinary averaging can lose architecture information [39, 53]. Aggregating only shared components also discards knowledge from components unique to a local architecture [75].

Solution 7a: Aggregate matching architecture parameters and model weights together. When clients use the same supernet, they can return both architecture parameters and model weights after local search. The server aggregates and redistributes them together [16, 14, 43]. This maintains one shared search state across rounds, but requires each variable to have the same meaning and shape at every client.

Solution 7b: Aggregate architecture encodings separately from model parameters. Clients can upload compact graph encodings without model weights. The server counts how often connections and operations occur and uses these frequencies to sample a global graph [39]. A related strategy assembles frequently recurring subgraphs directly [53]. Both preserve architecture information without matching all local model parameters. Compact encodings may contain fewer values than continuous gradients, but provide no formal privacy guarantee.

Solution 7c: Combine matching blocks based on distribution similarity. When different local architectures contain compatible blocks, the server can give more weight to blocks whose parameter distributions resemble that of the server block. Similarity is measured with Jensen–Shannon divergence [75]. This transfers parameters between matching blocks without assuming that the complete architectures match.

Solution 7d: Transfer knowledge between architectures through distillation. Client predictions or feature maps on shared inputs can teach a server supernet, sampled subnets or different blocks. A target architecture can therefore learn from clients that cannot run it or whose parameters do not match [75, 62, 34]. Compressed predictions can likewise support collaboration between entirely different architectures [30]. This avoids matching parameters directly, but requires shared inputs and transfers model behavior rather than complete parameters.

Challenge 8: Architecture Evaluation Depends on the FL Configuration

Because raw client data remain local, candidates cannot usually be trained and validated on the same pooled data [22]. Their scores instead depend on the FL setup, including the number of local epochs and the fraction of participating clients. An architecture selected under one setup may rank differently under another [43, 45, 23]. Changing these settings after search can therefore invalidate the earlier comparison.

Solution 8a: Search architectures and FL settings together. An alternating procedure can choose a training setup, search an architecture under it and use the result to choose the next setup [43]. Evolutionary and black-box strategies can instead search the architecture together with settings such as local epochs

and client participation, then evaluate each combination through federated training [45, 23]. Candidate scores then reflect the FL procedure in which the architecture will operate.

Candidate scores can also change with the current search state. Layer-wise updates may favor different operations at different depths, and an early preference for operations without parameters can prevent convergence [26]. Fixing one operation changes the computation graph and weight sharing, so the next choice may be evaluated before the remaining weights adapt [36]. Changing which subnets are trained can likewise make aggregated updates unstable and slow convergence [49]. A score can therefore reflect when and how a candidate was evaluated, not only its architecture.

Solution 8b: Regularize layer-wise architecture updates. During local differentiable search, an added term can align layer gradients with architecture parameters. This reduces depth-dependent preferences for particular operations before client updates are aggregated [26].

Solution 8c: Insert recovery rounds between architecture decisions. A progressive supernet search can first warm up the shared weights and then fix one operation at a time based on local validation accuracy. After each decision, several recovery rounds update the remaining weights before the next edge is evaluated [36]. Unlike Solution 2b, this pause helps weights adapt to each architecture change. Once all relevant edges are resolved, ordinary training continues with a fixed architecture.

Solution 8d: Vary which subnets are trained over time. The server can assign very different subnets in early rounds and increasingly similar ones as training advances. Early rounds explore separate parts of the search space, while later rounds concentrate on better understood regions [49]. These structured assignments also combine more consistently than independently sampled subnets.

5.4 Participation Dynamics

The participating clients and the timing of their updates can change throughout architecture search. Clients may disconnect, respond at different speeds, converge at different times or participate with different frequencies.

Challenge 9: Variable Client Participation Disrupts Search

Clients can become unavailable because of changing network conditions, battery limits or competing local work. A search that requires the same clients in every round may stall or lose updates for candidates or supernet paths assigned to disconnected clients [72].

Available clients may still complete search work at different speeds. A round may wait for the slowest client, while an update that arrives after the global search has advanced may already be outdated [20, 64, 5]. This differs from resource infeasibility, where a client cannot run the workload, and dropout, where it returns no update.

Solution 9a: Group split-model segments by training time. Split-model strategies can measure client computation and communication times and group segments that finish at similar times. Unlike Solution 6b, this groups clients only by observed timing so that supernet training waits less for the slowest segment [20].

Solution 9b: Adjust the influence of outdated search updates. The server can account for delay when adding a late update after the global search has advanced [64]. Differentiable search can instead give older updates less weight [5]. Unlike Solution 6a, these approaches adjust an update according to its age rather than how much each parameter was trained.

In personalized differentiable search, clients can learn architecture parameters at different rates because their data and local training differ. One shared stopping round can end some searches too early while allowing others to overfit, for example by increasingly favoring skip connections [34]. Clients can therefore need different stopping times even when their updates arrive promptly.

Solution 9c: Stop architecture updates independently. After a shared warm-up period, each client can stop updating its architecture when a local test detects growing preference for skip connections. Other clients continue until they meet their own stopping condition [34].

Partial participation and random client selection can cause some clients to contribute more often than others [72, 61]. Even if the search continues, frequently available clients influence shared weights and architecture rankings more often. The result may therefore poorly represent clients that participate less often.

Cross-cutting solution 6b. This solution can select participants by both resources and data coverage so that availability alone does not determine whose data enter the search [37, 29].

5.5 Search Security

NAS-in-FL communicates information about the architecture in addition to conventional model updates. Malicious participants can interfere with the search, while updates from honest clients may reveal private information.

Challenge 10: Architecture Search Introduces New Security Risks

NAS-in-FL creates additional attack targets because clients influence both model weights and search signals that determine the selected architecture. Malicious clients can manipulate either through poisoned data or local training, steering the search toward a model that supports a backdoor attack. Weight sharing can spread poisoned weights across candidates. Fourfold-scaled updates from one of eight clients degraded two discrete strategies, showing that poisoning can persist in the selected architecture [73].

Conditional observation. During differentiable supernet-based search, each candidate operation remains in a weighted mixture while clients update its weights and selection probability. [73] finds that an operation contributes less when its poisoned weights work against or contribute little to the main task.

[73] demonstrates this behavior only for the studied backdoor attacks and search spaces. Retraining the selected fixed architecture from scratch did not preserve its resistance to attack, suggesting that the effect belongs to the trained search state rather than the architecture itself.

Solution 10a: Pre-screen search-space modules on server data. The server can use its own data to select a compact set of accurate, efficient modules before federated search. In [73], this reduced complexity and limited overfitting to noise from backdoor triggers. Unlike Solution 1a, the mechanism tests modules using data. Unlike Solution 4b, it removes modules before search instead of changing their scores.

NAS-in-FL transmits data-dependent architecture gradients and rewards that fixed-architecture FL does not [46, 43]. Other strategies transmit structural encodings or personalized choices that may reveal characteristics of client data, individual examples or which client sent an update, although [39, 73] does not demonstrate such attacks. The risk depends on what is transmitted and the attacker's capabilities.

Solution 10b: Apply differential privacy to search signals. Differentiable strategies can limit the size of model and architecture gradients and add Gaussian noise before upload. Noise can also be added to rewards used for hyperparameter selection [46, 43]. In vertically partitioned search, the intermediate activations and backward gradients exchanged between parties can receive the same protection [33]. The noise can, however, reduce model and search performance [73].

Solution 10c: Apply secure aggregation to matching search variables. When every client updates the same differentiable supernet, model weights and architecture parameters have matching structures and can use the same secure-aggregation protocol [73]. This hides each client's updates from the server but requires every client to use the same search representation.

Cross-cutting solution 7b. The compact structural encoding exchanged in this solution may make it harder to reconstruct inputs because it has far fewer dimensions than the private data. The cited analysis

argues that a small derivative matrix can correspond to many image batches, making exact pixel recovery by that procedure impractical [39]. However, a compact encoding provides no formal privacy guarantee. It may still reveal properties of the client data or the source of an update and may require differential privacy or encryption. Table 5.1 summarizes the challenges and solutions. Cross-cutting entries indicate where a solution primarily discussed under another challenge also applies.

Table 5.1: Overview of the identified challenges, their primary solutions, and cross-cutting applications.

Theme	Challenge	Solution	Example publication
Client resource constraints	Challenge 1	Solution 1a	[57]
		Solution 1b	[9]
		Solution 1c	[54]
		Cross-cutting solution 7d	[62]
	Challenge 2	Solution 2a	[67]
		Solution 2b	[14]
		Solution 2c	[62]
		Solution 2d	[67]
		Solution 2e	[59]
		Solution 2f	[20]
		Solution 2g	[14]
	Challenge 3	Solution 3a	[77]
		Solution 3b	[5]
		Solution 3c	[29]
		Cross-cutting solution 4b	[76]
Client heterogeneity	Challenge 4	Solution 4a	[25]
		Solution 4b	[71]
	Challenge 5	Solution 5a	[40]
		Solution 5b	[25]
		Solution 5c	[18]
		Cross-cutting solution 6b	[29]
	Challenge 6	Solution 6a	[9]
		Solution 6b	[29]
		Cross-cutting solution 7d	[62]
Decentralized search state	Challenge 7	Solution 7a	[16]
		Solution 7b	[39]
		Solution 7c	[75]
		Solution 7d	[75]
	Challenge 8	Solution 8a	[43]
		Solution 8b	[26]
		Solution 8c	[36]
		Solution 8d	[49]
Participation dynamics	Challenge 9	Solution 9a	[20]
		Solution 9b	[64, 5]
		Solution 9c	[34]
		Cross-cutting solution 6b	[37]
Search security	Challenge 10	Solution 10a	[73]
		Solution 10b	[46]
		Solution 10c	[73]
		Cross-cutting solution 7b	[39]

6 Reusing Solution Mechanisms

A NAS-in-FL scenario is a tuple of a NAS method type and FL system characteristics. The NAS method type comprises its search strategy, search space and performance-estimation strategy. The FL system characteristics follow the dimensions defined in the background. Each row in Table 6.1 describes the broadest scenario range in which the solution mechanism can operate. “Any” includes every value defined for that characteristic. A parenthetical condition narrows this range only where the mechanism requires an additional property. These entries are conceptual compatibility judgments and do not claim that every included combination has been implemented or evaluated.

Table 6.1: Broadest possible NAS-in-FL scenario for which each solution is applicable.

Solution	Search strategy type	Search space type	Performance estimation strategy	Data heterogeneity	Data balance	Client computation resources	Network environment	Federation environment	Deployment environment	Trust and privacy requirements
S1a – Use resource-efficient search components	Any	Any (permits restriction to inexpensive components)	Any	Any	Any	Homogeneous or heterogeneous, with binding client limits	Any	Any	Any	Any
S1b – Send clients subnets tailored to their capacity	Supernet-Based Methods	Any (supernet-encodable and supports capacity-matched subnets)	One-Shot Models	Any	Any	Heterogeneous, with at least one constrained client tier	Any	Any	Any	Any
S1c – Search trainable adaptation modules over a frozen backbone	Any	Any (modular search over a transferable frozen backbone)	Any	Any (related target tasks)	Any	Homogeneous or heterogeneous, with binding client limits	Any	Any	Any	Any
S2a – Allocate candidate evaluation budgets adaptively	Black-Box Optimization Techniques	Any	Multi-Fidelity	Any	Any	Homogeneous or heterogeneous, with a constrained candidate-evaluation budget	Any	Any	Any	Any
S2b – Prune the supernet progressively	Supernet-Based Methods	Any (supernet-encodable and progressively prunable)	Any (must rank operations or subnets)	Any	Any	Homogeneous or heterogeneous, with a constrained search budget	Any	Any	Any	Any
S2c – Predict candidate performance	Any	Any (predictor-encodable)	Surrogate Models	Any (client-specific or collaborative prediction may be required under heterogeneity)	Any	Homogeneous or heterogeneous, with a constrained candidate-evaluation budget	Any	Any	Any	Any
S2d – Evaluate independent candidates or subnets in parallel	Black-Box Optimization Techniques and Non-Differentiable Supernet-Based Methods	Any	Any (independent candidate estimates)	Any	Any	Homogeneous or heterogeneous, with sufficient aggregate client capacity	Any	Stable or dynamic, with sufficient simultaneous availability	Any	Any
S2e – Reuse search-trained weights for candidate evaluation and deployment	Supernet-Based Methods	Any (supernet-encodable)	One-Shot Models and Weight Inheritance	Any	Any	Any	Any	Any	Any	Any

Continued on next page

Table 6.1 continued

Solution	Search strategy type	Search space type	Performance estimation strategy	Data heterogeneity	Data balance	Client computation resources	Network environment	Federation environment	Deployment environment	Trust and privacy requirements
S2f – Use split-training idle periods for local search	Any	Any (decomposable into locally searchable split-model segments)	Any	Any	Any	Homogeneous or heterogeneous, with sufficient capacity for local segment search	Any (split execution creates sequential idle periods)	Any	Any	Any
S2g – Jointly derive an architecture and its deployable weights	Supernet-Based Methods	Any (supernet-encodable)	One-Shot Models	Any	Any	Any	Any	Any	Any	Any
S3a – Identify known candidates with compact selectors or genotypes	Any (supports discrete candidate instantiation)	Any (compactly encodable and clients retain a compatible model)	One-Shot Models and Weight Inheritance	Any	Any	Homogeneous or heterogeneous, with capacity to retain the compatible model	Homogeneous or heterogeneous, with binding transfer limits	Any	Any	Any
S3b – Synchronize architecture parameters less frequently	Differentiable Supernet-Based Methods	Any (supernet-encodable)	One-Shot Models	Any	Any	Any	Homogeneous or heterogeneous, with binding transfer limits	Stable or dynamic, with repeated synchronized rounds	Any	Any
S3c – Aggregate search updates hierarchically	Any (produces aggregatable distributed feedback)	Any	Any	Any	Any	Any	Homogeneous or heterogeneous, with a hierarchical topology and constrained wide-area links	Stable or dynamic, with an intermediate aggregation tier	Any	Any
S4a – Train one elastic supernet for constraint-specific deployment	Supernet-Based Methods	Any (elastic supernet-encodable)	One-Shot Models	Any	Any	Heterogeneous	Any	Any	Heterogeneous	Any
S4b – Include deployment-resource costs in the architecture objective	Any (supports a resource-aware objective)	Any	Any (resource cost is observable or predictable)	Any	Any	Any	Any	Any	Uniform or heterogeneous, with an explicit deployment-resource budget	Any
S5a – Cluster clients by learned architecture preference	Differentiable Supernet-Based Methods	Any (shared architecture-parameter space)	One-Shot Models	Heterogeneous	Any	Any	Any	Any	Any	Any

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Table 6.1 continued

Solution	Search strategy type	Search space type	Performance estimation strategy	Data heterogeneity	Data balance	Client computation resources	Network environment	Federation environment	Deployment environment	Trust and privacy requirements
S5b – Optimize the worst-performing subnets across client partitions	Any (supports a worst-partition objective)	Any	Any (partition-specific performance evidence)	Heterogeneous	Any	Any	Any	Any	Any	Any
S5c – Factor the architecture into shared and task-specific components	Any	Any (factorable into shared and task-specific components)	Any	Any (clients solve related tasks requiring task-specific components)	Any	Any	Any	Any	Any	Any
S6a – Aggregate parameters according to their actual exposure	Supernet-Based Methods	Any (aligned shared weights with partial operation exposure)	One-Shot Models	Any	Any	Heterogeneous, with capacity-dependent operation or subnet eligibility	Any	Any	Any	Any
S6b – Form resource-feasible, data-representative client groups	Any	Any	Any (client-group-based performance evidence)	Heterogeneous	Any	Homogeneous or heterogeneous, with resource-feasible grouping	Homogeneous or heterogeneous, with latency or bandwidth constraints included in grouping	Stable or dynamic, with enough available clients to form representative groups	Any	Any
S7a – Aggregate aligned architecture and model variables together	Supernet-Based Methods (shared architecture state)	Any (aligned shared supernet)	One-Shot Models	Any	Any	Any	Any	Stable or dynamic, with an aligned participating search state	Any	Any
S7b – Exchange and aggregate discrete architecture encodings separately from model parameters	Black-Box Optimization Techniques and Non-Differentiable Supernet-Based Methods	Any (discretely encodable)	Any	Any (clients produce distinct architecture choices)	Any	Any	Any	Any	Any	Any

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Table 6.1 continued

Solution	Search strategy type	Search space type	Performance estimation strategy	Data heterogeneity	Data balance	Client computation resources	Network environment	Federation environment	Deployment environment	Trust and privacy requirements
S7c – Aggregate aligned blocks by parameter-distribution similarity	Any (supports shared or inherited compatible weights)	Any (contains structurally corresponding blocks)	One-Shot Models and Weight Inheritance	Any	Any	Any	Any	Any	Any (local architectures may differ)	Any
S7d – Transfer knowledge across architecture boundaries through distillation	Any	Any	Any	Any	Any	Any	Any	Any	Any	Data locality only or additional confidentiality or integrity requirements, with permissible common distillation inputs
S8a – Search architectures and FL control variables jointly	Any (architecture and FL variables are jointly optimizable)	Any (extended with FL control variables)	Any	Any	Any	Any	Any	Stable or dynamic, with the relevant participation variables included in the search	Any	Any
S8b – Regularize layer-wise architecture updates	Differentiable Supernet-Based Methods	Any (shared supernet)	One-Shot Models	Heterogeneous	Any	Any	Any	Any	Uniform (one shared architecture)	Any
S8c – Separate structural decisions with recovery rounds	Supernet-Based Methods and Weight-Inheriting Black-Box Optimization	Any	One-Shot Models and Weight Inheritance	Any	Any	Homogeneous or heterogeneous, with sufficient capacity for recovery training	Any	Stable across recovery rounds	Any	Any
S8d – Vary subnet coverage over the search	Supernet-Based Methods	Any (mask- or path-encodable supernet)	One-Shot Models	Any	Any	Heterogeneous	Any	Stable or dynamic, with repeated client sampling	Any	Any
S9a – Group split-model segments by training latency	Any	Any (decomposable into split-model segments)	Any	Any	Any	Heterogeneous	Heterogeneous	Any	Any	Any
S9b – Compensate or down-weight stale search updates	Any (iterative search accepts late feedback)	Any	Any	Any	Any	Any	Heterogeneous latency	Stable or dynamic, with late updates	Any	Any

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Table 6.1 continued

Solution	Search strategy type	Search space type	Performance estimation strategy	Data heterogeneity	Data balance	Client computation resources	Network environment	Federation environment	Deployment environment	Trust and privacy requirements
S9c – Stop architecture updates independently	Any (supports client-specific stopping and selection)	Any	Any	Heterogeneous	Any	Any	Any	Stable or dynamic	Any	Any
S10a – Pre-screen search-space modules on server data	Any	Any (contains server-screenable modules or operations)	Any	Any	Any	Any	Any	Any	Any	Additional integrity requirements and representative server validation data
S10b – Apply differential privacy to search signals	Any (communicated search signals are protectable)	Any	Any	Any	Any	Any	Any	Any	Any	Additional confidentiality requirements
S10c – Securely aggregate aligned search variables	Any (uses aligned aggregatable numerical search state)	Any (aligned search variables)	Any	Any	Any	Any	Any	Stable or dynamic, with a secure-aggregation-compatible cohort	Uniform (one shared architecture state)	Additional confidentiality requirements

7 Discussion

- the further the FL system deviates from centralised NAS, the more challenging it is => conversely, when NAS-in-FL degenerates to centralised NAS in some aspects it becomes simpler - show how some solutions limit compatibility with others and how they are - the search space trade-off - suprisingly many NAS-in-FL methods do X - we expected federation scale to play a more prominent role, but this was barely ever considered by NAS-in-FL methods and the largest federation scale currently in the literature is ..., signifcantly smaller than regular FL setups - similar to NAS research, supernet-based methods have become more popular over time - performance evaluation is basically universally just training a candidate architecture for a couple of epochs, making the main differentiating factors for challenge applicability the NAS search strategy type and secondarily the search space - The FL system characteristics which NAS-in-FL methods apply to tend to be underspecified

7.1 Principal Findings

7.2 Implications for Practice and Research

- clear guidance on existing related solutions and challenges and the context/circumstances under which challenges arise - potentil for re-usable solutions to challenges

7.3 Limitations

- rely on literature for facts, i.e. described only challenges and solutions from the literature - few NAS-in-FL methods establish an explicit causal link between FL system characteristics and challenges, therefore this needs to be treated as corrolation evidence, not causal

7.4 Future Research

- large share

This chapter interprets the challenge–solution relationships, explains their implications for reusing solution mechanisms, and qualifies the conclusions supported by the reviewed literature.

7.5 Principal Findings

The analysis identified ten challenges across five themes and 34 solution mechanisms. Tables 5.1 and 6.1 show that these mechanisms can be reused when the target NAS-in-FL scenario satisfies their prerequisites. Their reuse requires matching the mechanism to the FL system characteristics, the NAS design, and the other mechanisms in the configuration.

Challenge relevance depends on which search functions rely on clients. Client-side search workloads expose the process to resource constraints, distributed candidate evaluation makes evidence dependent on local data, and decentralized search-state updates require coordination across participants. Retaining a function at the server can avoid the corresponding client-side constraint, but may require representative

server data or greater trust in the server. The challenges consequently arise from specific distributed dependencies rather than from decentralization as one uniform property.

Solution mechanisms also interact. Assigning capacity-compatible subnets reduces client workloads, but causes operations available only to capable clients to receive fewer updates. Exposure-aware aggregation is then needed to prevent unequal training from distorting the shared state [9, 65, 4]. Similarly, specialized architectures can accommodate different deployment requirements while removing the parameter alignment required for ordinary aggregation. Structural aggregation or distillation can restore collaboration under this condition [39, 75, 62]. A solution should therefore be assessed as part of a configuration rather than as an independent remedy.

The search space creates a recurring trade-off between architectural coverage and coordination. Restricting it to inexpensive, structurally aligned components lowers client workloads and enables direct aggregation, but may exclude architectures that better fit particular data or devices. Elastic or personalized spaces cover more deployment conditions, but increase search effort and can produce unequal parameter exposure or incompatible local states. The search space thus serves both as the set of possible architectures and as a coordination contract between clients.

Brief training is the predominant performance-estimation strategy in the reviewed literature. This common choice means that challenge applicability is often determined more directly by the workload assigned to clients, the relationship between their search states, and whether the search produces a shared or specialized architecture. Broad search-strategy categories alone can conceal these differences. Reasoning about reuse therefore requires the more specific scenario properties retained in Chapter 6.

7.6 Implications for Practice and Research

The synthesis provides a diagnostic process for selecting solution mechanisms. Practitioners can first characterize the intended FL system and NAS design, identify the challenges activated by this combination, and then select mechanisms whose prerequisites are satisfied. They must subsequently check whether the chosen mechanisms introduce additional costs or incompatible representations. The tables support this process, but they neither rank mechanisms nor guarantee that combinations not evaluated together will be effective.

For research, the synthesis complements method-level taxonomies by treating challenges and solution mechanisms as separate units. This exposes relationships that remain hidden when several mechanisms are presented as one NAS-in-FL method. It also provides a common vocabulary for comparing mechanisms under stated conditions. Future publications can support such comparisons by reporting the FL system characteristics, client workloads, exchanged search state, search space, and performance-estimation strategy used in their evaluations.

7.7 Limitations

The review depends on what the selected publications report. It can include only challenges, solution mechanisms, and applicability conditions described with enough detail to meet the inclusion criteria. Operational problems that have not been published or that receive only general descriptions may therefore be absent. The Scopus search and citation searching reduce this risk but do not establish that the corpus covers every relevant publication.

The thematic analysis also requires judgment when concepts are named, separated, merged, and mapped [3]. Explicit definitions and an audit trail make these decisions transparent, but do not eliminate interpretive subjectivity. Moreover, few publications isolate the effect of one FL characteristic or solution mechanism. The resulting mappings represent reported relationships and mechanism-based applicability

judgments, not independent estimates of causal effects or comparative effectiveness.

Evaluations differ in their datasets, search spaces, client populations, participation, resource models, and training budgets. These differences prevent a quantitative ranking of the mechanisms. Incomplete descriptions of FL systems further limit transfer judgments because an unreported condition cannot be distinguished from one that was absent. Publication frequency therefore indicates attention within the corpus, not practical prevalence, severity, or solution quality.

7.8 Future Research

Future evaluations should test individual mechanisms and selected combinations under shared configurations. Varying one FL condition or NAS property at a time would provide stronger evidence for the relationships identified in this review. Such evaluations should separate search computation, final-model training, communication, wall-clock time, candidate-ranking fidelity, and final architecture quality.

Scale and participation require particular attention. Experiments should vary the number of clients together with partial participation, dropout, latency, and capability-dependent selection, since a larger simulated population alone does not reproduce cross-device dynamics. Publications should also distinguish costs measured on client hardware from those simulated on centralized infrastructure.

A shared reporting scheme would make results easier to compare. At minimum, publications should describe client resources, network conditions, data distribution, participation, federation size, trust assumptions, client search workloads, exchanged search state, search space, and final-model development. Research should also examine sparsely studied areas of the challenge–solution map and test whether low-cost performance estimators from centralized NAS remain reliable when candidate evidence is distributed across heterogeneous client data [17].

8 Conclusion

- the amount of possible knobs to turn in NAS-in-FL is astronmical and the literature has only investigated a small subset of them - NAS in FL trails behind NAS and FL - provide some clear direction for future research

Abbreviations

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